

J40 cooling-pack refurbishment and fabrication

Actual-part references, measurement capture, recoring and restoration steps, fabricated mounting interfaces, finish controls and budget

ACTUAL RETAINED R0	ACTUAL RETAINED C0
	
Removed copper/brass radiator. The old core is not the dimensional authority for a replacement until the radiator shop strips and measures the tanks and headers.	Removed full-face condenser. The complete installed envelope includes brackets, pipes, ports and the receiver-drier, not just the fin field.

Configuration

Toyota 2H naturally aspirated. No turbocharger, intercooler/K0, charge-air pipework or second small front fan. The finished pack has one central FS front A/C pusher and one FL rear radiator puller/shroud.

Dimensional status

The project record contains no released actual R0, C0, G0, FS, FL, chassis-interface, arm or holder dimensions. Rev R therefore removes the former planning estimates. Blank ACTUAL fields below must be completed from the stripped parts and vehicle before any final cutting, drilling, welding, core order or hose order.

Technical authority	Fabrication state	Issue date
Rev P naturally aspirated cooling-pack guide and Rev P handoff package	MEASURED STRUCTURAL RELEASE HOLD	16 Aug 2026

1. What is controlled now

These are acceptance requirements or conditional starting rules. They are not measurements of the retained parts.

Control	Numeric requirement	Meaning / fabrication consequence
Direct centreline	G0, FS and C0 each within 2 mm of VCL	Measure each independently from the fixed chassis master datum. Do not derive one component centre from another.
Local alignment	G0-to-fixed grille aperture and FS-to-C0 each within 2 mm in X and Z	Separate local checks; they do not replace the direct VCL readings and cannot be tolerance-stacked.
Measurement capability	0.5 mm repeatability or better recommended	The method must resolve a 2 mm acceptance limit. Record instrument, setup and three readings where practical.
Static proof	Total 2 x MR for 10 minutes	Distribute at the two S0 saddle centres in the released loaded-radiator distribution. This is not road-load/fatigue design release.
X0 trial stock only	40 x 40 x 3 mm mild-steel SHS only if L0 is no more than 700 mm and MR no more than 35 kg	Permitted only as a mock-up/starting rule. Final section, joints and welds require structural calculation and signed drawing.
FL moving clearance	At least 20 mm radiator-to-blade static and 15 mm blade-to-shroud radial	Functional minimums to verify on the real installed assembly; they do not define shroud or carrier dimensions.

Actual dimensions still required

Assembly	Missing actual record	Do not release until
R0 radiator	Complete W/H/D; active face; core pack; tank/header joint; necks; cap seat; drain; rails; every ear/hole; lower locators	R0-H and S0/B0 sheets are signed and the radiator shop has stripped/measured the sample
C0 condenser	Fin field; complete W/H/D; drier; brackets; pipes; ports; tool and removal sweep	C0 map and specialist retain/replace decision are signed
G0 guard	Perimeter frame W/H/D; diagonals; sound contact lengths; holes; mesh clearance; removal vector	G0-H rigid template and CL0 readings are signed
FS / FL	Rotor axes; complete frames/tabs/guards; motors; plugs; cable bends; current; installed depth	Electrical/airflow tests and component maps pass
Vehicle / mounts	VCL origin; connector faces and holes; L0; MR; S0/B0; all obstructions and service sweeps	D00, A0-D-L/R, S0, B0, L0/MR and VEH records are signed

No photo scaling

Tape-in-frame photographs identify measurement locations and condition only. Perspective, tape angle and hidden hard points make them unsuitable for a cut list or replacement-core order.

2. Actual R0 radiator - condition and restoration

Retain the original copper/brass hard parts only where the radiator shop finds them sound; replace the tired heat-transfer core.

ACTUAL CORE DAMAGE	ACTUAL LOWER-EDGE DETAIL
	
Dirt, flattened fins and local damage. This is evidence for recoring and shop testing, not a cosmetic-only repaint.	Historical added support/leg and lower area. Classify every added piece on R0-H before removal, repair or rejection; do not reproduce it automatically.

Step	Radiator-shop instruction	Evidence / pass
R0-1	Photograph and tag all faces; record cold pressure, flow restriction, leaks, damaged fins, seams, neck, drain, rails, ears and locators.	Signed incoming condition sheet
R0-2	Build a rigid jig from the sound tank/header/rail relationships before unsoldering. Record complete actual dimensions and R0-H/S0/B0 interfaces.	Dated jig record and actual measurement sheet
R0-3	Desolder and remove the old core without twisting tanks or rails. Chemically clean using a copper/brass-and-solder-compatible inhibited process.	Bare parts inspected; no thinning, cracks or blocked passages
R0-4	Repair sound tanks, headers, filler/cap neck, drain, rails and approved ears. Any R0-E hot work is radiator-shop-only and occurs before final tests.	Retain/repair/reject decision for each hard part
R0-5	Order the new copper/brass core from the stripped physical sample and stated naturally aspirated duty. Record tube/fin construction and finished core dimensions supplied by the shop.	Core purchase record tied to R0 serial/tag
R0-6	Jig, solder and square the assembly without loading tanks/core through future brackets. Pressure-test and flow-test after all hot work.	Written before/after pressure and flow results
R0-7	Apply only a thin radiator-shop heat-exchanger coating; keep fins open and cap seat, necks, drain, threads, earths and rubber lands clean.	Straight/open fins, clean interfaces and final dimensions recorded

3. Actual C0 condenser and receiver-drier

The large canister is the existing receiver-drier. It supplies the geometry reference only; install a new sealed compatible unit at final assembly.

ACTUAL C0 COMPLETE ASSEMBLY	ACTUAL RECEIVER-DRIER LOCATION
 Measure the complete condenser including pipes, brackets, ports and tool sweep. The fin field alone does not define the installed envelope.	 Tall black can beside C0. It filters debris, absorbs moisture and provides a liquid-refrigerant reserve. Keep the old unit only as a port, bracket and routing sample.

Step	A/C specialist instruction	Evidence / pass
C0-1	Cap ports, tag every pipe/bracket and record actual fin-field and complete installed envelope dimensions before disassembly.	C0 map with drier/port/tool/removal envelopes
C0-2	Use neutral aluminium-safe cleaner, low pressure and a correct fin comb. Do not use harsh brightener or a pressure jet.	Clean, open fins with no new tube damage
C0-3	Specialist leak/pressure and contamination assessment using the specified safe test medium and pressure for the identified condenser.	Written retain/repair/replace decision
C0-4	Retain sound C0 only if it passes. Otherwise commission a matched parallel-flow replacement from the complete physical envelope and port map.	Replacement drawing/sample comparison if required
C0-5	Fit a new sealed compatible receiver-drier, new HNBR O-rings/seals/caps and the identified refrigerant oil during final specialist assembly.	Drier part ID, oil/refrigerant identity and evacuation record
C0-6	Mount C0 on independent C1 brackets with isolators and shoulder/crush sleeves; preserve pipe and tool sweep.	No load transferred to R0, G0 or another heat exchanger

4. Actual G0 guard, FS pusher and FL puller

Restore and test the retained parts first. Replace a fan only if the complete tested assembly cannot meet the measured installation.

ACTUAL G0	ACTUAL FS	ACTUAL FL
 Removable expanded-mesh guard and frame. Hold only the perimeter frame; never clamp the mesh.	 The one central front A/C pusher candidate. The complete frame/rotor datum, not blade diameter alone, controls F1.	 Large rear puller and shroud. F2 supports the shroud independently and seals to the sound R0 frame.

Part	Restoration / test	Fabrication interface
G0	Jig; clean; straighten/repair perimeter; repair or replace damaged mesh while preserving open area; deburr; apply thin compatible satin/semi-gloss finish and edge trim.	G1 lower cradles and G2 accessible keepers contact broad, sound perimeter lengths through EPDM; preserve drainage and a documented removal vector.
FS	Record maker/label, polarity, rotation, push direction, start/run current, voltage drop, bearing noise, wobble, rub, plug/cable and installed airflow before retain decision.	One independent F1 carrier. Complete frame/rotor datum is measured directly to VCL and separately to C0. Own relay, fuse and earth.
FL	Record maker/label, polarity, pull direction, start/run current, bearing/rub, complete shroud, plug/cable bend, seal land and installed airflow.	Independent F2 carrier and replaceable EPDM seal to the sound R0 frame; maintain moving clearances and zero tank/core load. Own relay, fuse and earth.

Alignment

The complete repaired G0 perimeter-frame centre, complete mounted FS frame/rotor datum and C0 usable-fin-field centre each require their own direct reading to VCL within 2 mm. Then record G0-to-fixed-aperture and FS-to-C0 X/Z offsets separately within 2 mm.

5. Where to take the actual measurements

Create D00 first from fixed chassis references; every coordinate and centre offset is then measured directly from it.

ACTUAL VEHICLE CONTEXT	ACTUAL COMPONENT DATUM
 Record the intended structural connector bearing face and hole group on each side. Do not scale this photograph and do not use steering, suspension, bumper, tow or thin sheet as a pickup.	 Use hard, repeatable rails, tank/header edges, neck axes, locator axes and sound ear planes. Record complete hard points and service sweeps.

Record	Measurement points / method	Actual result
D00	VCL plus fixed chassis X/Y/Z origin, projected at G0, FS, C0 and A0 planes. Record origin marks, instrument, setup and repeatability.	Origin: _____ Method: _____ Repeatability: ____ mm
A0-D-L	Left connector bearing face XYZ; every hole centre/diameter from two orthogonal edges; local section/thickness; bolt/nut/tool/removal sweep.	Template ID: _____ Face XYZ: _____ Holes: _____
A0-D-R	Right connector bearing face and hole group by the same method; do not mirror-assume.	Template ID: _____ Face XYZ: _____ Holes: _____
L0	Clear span between released A0/A1 end-envelope limit planes at X0 level, including joint/tool/drainage envelopes.	L0 = _____ mm
MR	Finished, capped R0 filled to installed level, including retained fittings; record scale and uncertainty.	MR = _____ kg Scale: _____ Uncertainty: _____
S0-L/R	R0 lower locator centres/diameters plus final R1 saddle centres, cup depths and installed loaded heights.	L XYZ/dia/height: _____ R: _____
B0-L/R	With R0 naturally seated and sharing weight across both R1 saddles, transfer sound upper keeper axes/planes to fixed structure.	L XYZ/plane: _____ R: _____

6. Component and alignment measurement sheet

Complete every blank from the actual repaired part, including all brackets, plugs, pipes, cable bends and removal/tool envelopes.

Record	Required actual readings	Actual / template / date
R0 complete	W/H/D; active face W/H; core pack; body-to-cap height; upper/lower/overflow neck OD/axis; drain; rails; each R0-H ear/hole/plane; lower locators	W ____ H ____ D ____ mm; map/template ____
C0 complete	Fin field W/H/centre; body W/H/D; complete W/H/D including drier, pipes, ports and brackets; tool and removal sweep	Fin ____ x ____ mm; total ____ x ____ x ____ mm; map ____
G0-H	Frame W top/mid/bottom; H left/centre/right; maximum D; both diagonals; sound contact lengths/planes/holes; mesh-to-fixed-grille clearance; removal vector	W ____/____/____; H ____/____/____; D ____; diag ____/____ mm
FS	Rotor diameter/axis; complete frame W/H/D; tabs/guard; motor/plug/cable bend; current/voltage/airflow/direction	Rotor ____; frame ____ x ____ x ____ mm; test sheet ____
FL	Rotor diameter/axis; shroud W/H/D; tabs; motor/plug/cable bend; seal land; static/radial moving clearances; current/airflow/direction	Rotor ____; shroud ____ x ____ x ____ mm; clearances ____/____ mm
CL0 direct	Signed lateral offset from D00-VCL to G0 frame centre, FS frame/rotor datum and C0 fin-field centre, each independently	G0 ____ mm FS ____ mm C0 ____ mm; each PASS []
CL0 local	Signed X/Z G0 frame-to-fixed-aperture and FS frame/rotor-to-C0 fin field; raw left/right/top/bottom readings retained	G0 X/Z ____/____ mm; FS X/Z ____/____ mm; PASS []
VEH	Fixed grille, bonnet/latch, bumper, steering, suspension, tow, support steel, engine movement, hoses, pipes, wiring, tools and independent removal paths	Envelope drawing/template ID ____

Measurement release

Fabricator and owner sign the common-datum coordinate sheet and rigid templates. Record raw readings, not only calculated centres. Left and right parts are measured separately; neither side is assumed symmetric.

Measurement sign-off

Role	Name	Instrument / template IDs	Signature / date
Measurer			
Fabricator			
Owner witness			

7. Fabrication reference schedule

Each part is drawn from its named measured interface. Rev R contains no bracket cut size, hole size, material thickness, weld size or torque release.

Ref / qty	Function and attachment	Required inputs	Issued fabrication evidence
X0 / 1	Full-width lower crossmember between A0/A1 ends; carries X1 saddle seats	L0, MR, A0-D, S0, road/shock/fatigue loads, drainage and service	Signed structural plan/elevation/sections, section, joints, welds, finish
X1-L/R / 2	R1 saddle seats on X0; carry all filled-R0 vertical weight	S0, selected R1 cup/base/loaded height, locator map and drainage	Seat/gusset details and weld/inspection plan
A0-L/R + A1-L/R / 2 assys	Short arms and connector ends mating directly to actual chassis connector bearing faces	A0-D-L/R rigid templates, L0, X0 interface, local parent section, tool/removal envelope	Separate LH/RH drawings; material, cut, bends, holes, edge distance, weld/bolts and torque
G1-L/R / 2	Broad EPDM-faced lower cradles on sound G0 perimeter frame only	G0-H contact lengths/planes, G0 centre, CL0, drainage and withdrawal path	Blank/bend/section, sleeve-controlled hardware and finish
G2-L/R / 2	Accessible upper/side positive keepers on G0 frame only	G0-H, final G1 seating, CL0 and removal/tool sweep	Keeper/escape-lip and anti-chafe detail
R0-E / as required	Radiator-side ear/tab repair or replacement on sound rail/header only	R0-H classification, jig and tank/core clearance	Radiator-shop drawing/process; pressure/flow test after hot work
R3-U / 2	Upper locating keepers; restraint only and neutral in vertical load	B0 after final R1/X1 seating, R0-H and EPDM/crush-sleeve stack	Keeper drawing, compression control and neutral-load witness method
R3-L / 0 or 2	Optional side stabilisers; zero vertical load	Dry-fit justification, R0-H sound interfaces and service sweep	Issued only if required; otherwise N/A
C1 / measured	Independent removable C0 brackets with isolation	Complete C0 map, ports/pipes/drier/tool sweep and C0-to-VCL	Bracket and shoulder/crush-sleeve joint drawings
F1 / 1 set	Independent central FS carrier upstream of C0	Complete FS map, current/airflow, FS-to-VCL and FS-to-C0	Carrier, guard/plug/cable clearance and removable joint drawing
F2 / 1 set	Independent FL shroud support and R0 frame seal land	FL and R0-H maps, moving clearances, engine/service envelope	Frame, EPDM seal compression and zero-core-load detail
E1 / 1	Splash-resistant removable relay/fuse carrier	FS/FL currents, cable runs, earths, drainage and service access	Electrical layout and enclosure/carrier drawing

Drawing rule

Every issued drawing carries a unique revision, quantity, material and standard, dimensions/tolerances, hole/bend/weld definition, fastener and torque, finish, interface reference and inspection point. This PDF is the scope and measurement record, not the cut drawing.

8. Lower cradle and short connector arms

The retained arms are loose and unattached: shorten or replace them on the bench so the finished pair ends at the highest required interface with no unused upright projection.

ACTUAL INTERFACE CONTEXT - VERIFY ON VEHICLE



Historical front-support view only. Resolve the true current connector bearing planes and hole groups with A0-D-L/R templates before design. No photograph establishes the finished arm height.

Sequence	Fabrication instruction	Hold / inspection
A	Identify and inspect each loose arm blank. Record grade evidence, section, wall/plate thickness, straightness, corrosion, old holes, cracks and previous heat damage.	Replace with certified matched steel if grade/condition/weldability is not released
B	Template the actual LH/RH connector bearing faces and hole groups separately. Model bolt head, nut, socket and removal clearance.	No mirror assumption; no new chassis holes, slots, reaming or forced alignment
C	Set final R1/X1/R0 seating, B0 keeper planes and X0 level. Derive each arm height between its connector bearing plane, X0 interface and highest assigned functional interface.	No unused upper projection and no tank/core load
D	Competent person designs X0/X1/A0/A1 for vertical, fore/aft, lateral, shock, vibration and fatigue loads, plus welds, bolts, bearing and corrosion/drainage.	The 2 x MR proof does not replace calculation
E	Make a tack/temporary steel or rigid template assembly; dry-fit actual saddles, rubbers, fasteners and R0. Correct the drawing, then fabricate and inspect the released parts.	Final dimensions come only from the signed drawing

Conditional X0 starting rule

40 x 40 x 3 mm mild-steel SHS is permitted only as mock-up/starting stock when measured L0 is no more than 700 mm and measured MR is no more than 35 kg. It is not a final structural selection.

9. Guard, radiator and fan-holder fabrication

Holder geometry follows repaired-part templates. Use broad rubber-faced contacts, controlled compression, drainage and independent removal.

G0-H SOURCE	R0-H SOURCE
	
Measure sound straight perimeter-frame lengths and planes for G1/G2. Mesh is not a bearing surface.	Classify rails, headers, ears and locators. R0-E/R3/F2 may use only sound released interfaces and never tank/core/fin surfaces.

Holder	Specific construction requirement	Acceptance
G1	Two lower frame cradles with broad replaceable EPDM contact, positive lateral control, open drainage and no mesh contact.	G0 rests without twist; direct G0-to-VCL and local aperture offsets pass
G2	Two accessible upper/side keepers with sleeve-controlled EPDM compression and a positive escape lip/retention feature.	No preload distortion; G0 removes by the documented vector without disturbing C0/FS
R3-U	Two short keepers at B0 using new EPDM and fitted crush sleeves. Tightening locates/restrains only.	Both R1 saddles remain loaded; documented post-torque neutral-load witness
R3-L	Add only if dry fit proves side restraint is required; use a sound R0-H interface and no vertical support.	N/A or zero vertical load recorded
F1	One central FS carrier with complete-frame tabs, guard, motor, plug and cable bends represented. Independent of G0/C0/R0.	Direct FS-to-VCL and local FS-to-C0 offsets pass; serviceable fasteners
F2	Carrier supports the full FL shroud and replaceable seal land; no through-core ties or tank/core loads.	Shroud seals to sound R0 frame; moving/engine/tool clearances pass

10. Ten-stage restoration and fabrication sequence

Each stage ends with named evidence. Do not paint or order final hoses until the bare-metal full-stack dry fit passes.

Stage	Work	Release evidence
1. Inventory	Tag and photograph R0/C0/G0/FS/FL, loose arms and all brackets. Cap ports. Baseline-test R0, C0 and both fans.	Signed retain / repair / replace decisions
2. Measure	Establish D00; capture A0-D, B0, S0, CL0, G0-H, R0-H, L0/MR, complete component and vehicle envelopes with rigid templates.	Signed raw readings, common-datum sheet and templates
3. R0	Jig, strip, clean, repair, order/fix core by sample, complete approved R0-E hot work, pressure/flow test and thin coat.	Radiator-shop job sheet and final measured R0 map
4. C0	Neutral clean, fin-comb, specialist leak/contamination test; retain or replace by complete sample; new drier/HNBR at assembly.	A/C specialist result and final C0 map
5. G0/fans	Repair G0 frame/mesh; test FS and FL identity, direction, current, bearings, complete frame, wiring and installed airflow.	G0-H and fan test sheets
6. Structure	Engineer and fabricate X0/X1/A0/A1 to actual connector, saddle and load records.	Signed structural drawings and weld/fastener inspection
7. Holders	Fabricate G1/G2/R3/C1/F1/F2/E1 from their named actual templates with sleeved EPDM isolation.	Tagged parts, drawings and bench inspection
8. Bare dry fit	Assemble the opaque full stack with actual rubbers, hardware, pipes, plugs, cable bends, hoses and tools. Record direct and local CL0 values.	No forced alignment; all clearances/removal paths and centre checks pass
9. Proof/finish	Apply released total 2 x MR / 10 minute proof, reinspect datums and joints, then deburr, clean, prime, finish and cavity-protect eligible steel.	No set/crack/movement; coating and drainage inspection
10. Assemble/test	Fit cooling/A/C plumbing and two independent fan circuits; pressure/evacuate/charge, heat-cycle, hot-idle A/C, road/load and reinspection.	No leak/rub/boil/purge/thermal escalation or witness movement

Stop-work rule

If a retained interface is cracked, thinned, distorted, unidentifiable or would force misalignment, stop and revise the drawing or replace the part. Do not solve fit with slots, reaming, loose washer stacks, self-tappers or through-core ties.

11. Proof, finishing and final assembly

Proof the accepted bare-metal structure before coating. Keep heat-exchanger, rubber and electrical interfaces free of unsuitable finish.

Operation	Specific method	Pass / record
Dry fit	Fit all actual components, rubbers, sleeves, fasteners, pipes, plugs, hoses and tool envelopes. Check bonnet/latch, grille, bumper, steering, suspension, tow and engine movement.	Independent removal paths; no contact, forced pull or distorted rubber
Static proof	Apply a total static load of 2 x MR, distributed between S0 saddle centres in the released loaded-radiator distribution for 10 minutes using a fixture that avoids point loading.	No permanent set, crack, connector/fastener movement, saddle displacement, arm distortion or alignment loss
Post-proof	Repeat A0-D/S0/B0/CL0 critical readings, inspect welds/bolts/connectors and verify R3-U remains neutral while both saddles carry weight.	Signed before/after sheet and witness photographs
Steel preparation	Deburr/radius; degrease; local ferrous rust treatment within one compatible system; preserve drains, bearing faces, earths, threads and rubber seats.	Clean, keyed, dry substrate and open drains
Steel finish	Professional 2K epoxy primer plus compatible 2K polyurethane satin/semi-gloss; cavity wax only after cure in eligible closed sections.	Continuous thin finish, correct cure, no blocked drainage or coated interfaces
Heat exchangers	R0 gets thin radiator-shop coating only; C0 fins stay open and clean. Do not powder coat, fill or thick-prime cores.	Open fins and readable labels/ports
Final hardware	Use released class/grade, broad washers, positive/prevaling locks, grip length and fitted crush/shoulder sleeves; torque and witness-mark as drawn.	Torque sheet, sleeve-controlled rubber compression and visible witness marks

Final installation checks

- Use new measured coolant hoses, overflow hose, constant-tension clamps, approved cap, Toyota 2H thermostat/seal and one compatible long-life coolant family.
- Use a new sealed compatible receiver-drier, HNBR seals/caps and the identified refrigerant oil; evacuation and charging are A/C-specialist work.
- Give FS and FL separate sealed relays, fuses and earths sized from measured start/run current and route length; record voltage drop and connector temperature.
- Heat-cycle and bleed; hot-idle with A/C; road/load test the naturally aspirated vehicle; reinspect after cool-down and the first loaded journey.

12. Parts, chemicals and consumables

Buy wear, sealing and service parts new. Retain major assemblies only after the tests above; fabricate every holder from the signed actual interface drawings.

Group	Required parts / consumables	Control
Radiator	Shop-sourced copper/brass core, compatible solder/flux, thin heat-exchanger coating, R1 saddles, approved cap, 2H thermostat/seal, measured moulded hoses, overflow hose, constant-tension clamps and coolant	Core, necks and hoses from stripped/final R0; no catalogue guess
A/C	New compatible receiver-drier, HNBR O-rings/seals/caps and identified refrigerant oil	Match final C0 and refrigerant system; never reuse old drier
Mounting	Automotive EPDM strips/bushes, fitted crush/shoulder sleeves, edge trim, shroud seal, released plated fasteners, broad washers and positive/prevaling locknuts	Rubber compression is sleeve-controlled; no loose spacer stacks
Electrical	Two separate sealed relay/fuse branches, measured cable, terminals/connectors, earth hardware, heat-shrink, grommets, loom and P-clips	Size after actual FS/FL current and route-length tests
Cleaning	Water-based degreaser; copper/brass-and-solder-compatible inhibited radiator cleaner; neutral aluminium-safe condenser cleaner; 70-90% IPA for compatible bracket wipe	Follow current SDS; cap ports; no motor immersion, harsh acid/caustic or pressure jet
Steel coating	One compatible 2K epoxy and 2K polyurethane satin/semi-gloss system; local phosphoric steel prep; compatible cavity wax after cure	Professional controls/PPE; isolate from aluminium, copper/brass, solder, cores, rubber, earths and threads
Assembly	EPDM-safe assembly aid and light dielectric grease for connector seal lips/boots	No petroleum grease on EPDM and no dielectric grease between contact faces

Specialist safety

Use weld consumables matched to verified parent steel and an approved WPS. Never weld near chlorinated-cleaner residue, pressure-test A/C with oxygen, vent refrigerant or spray 2K products without the specified professional respiratory and workshop controls.

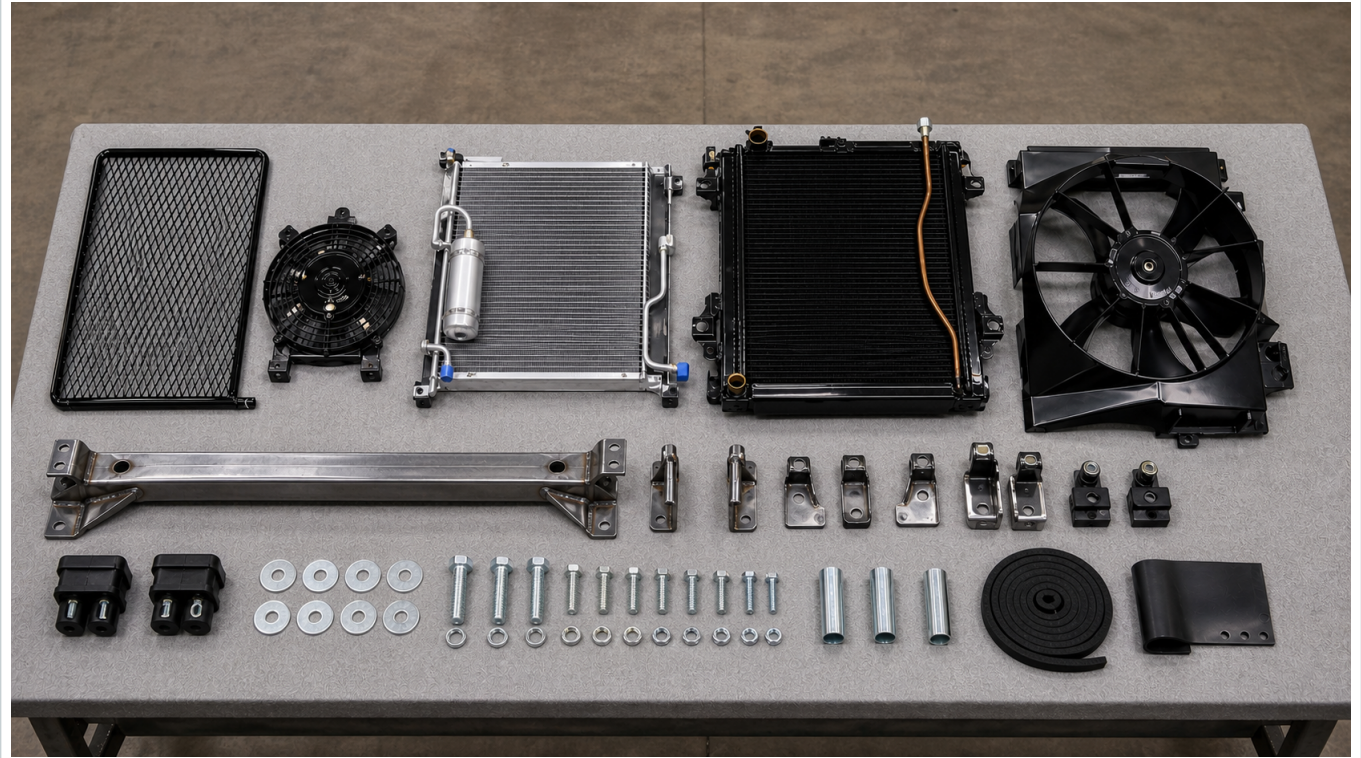
Quality finish target

Straight, open fins; neat and fully cleaned solder work; satin/semi-gloss coated steel; thin heat-exchanger coating; clean plated hardware; replaceable black EPDM; visible drainage; accessible service fasteners; labelled wiring and capped ports.

13. Example renewed finish - appearance only

This single example image communicates cleanliness and organisation. It is deliberately excluded from every dimensional and fabrication decision.

EXAMPLE FINISH ONLY - ILLUSTRATIVE AND NON-DIMENSIONAL



A tidy bench presentation of restored heat exchangers, guard, one FS, one FL, fabricated supports, isolators and hardware. The rendered drier finish, bracket shapes, hole patterns, arm proportions, pipe routes and component spacing are not evidence of the actual parts and must not be copied.

Use this image for	Do not use this image for
Finish quality; part cleanliness; organised hardware; replaceable rubber; serviceable presentation	Dimensions; cut lengths; material thickness; hole locations; drier proportions; brackets; pipes; wiring; fan spacing; centreline proof

Fabrication evidence hierarchy

1) Actual tagged component and vehicle. 2) Signed D00/common-datum readings and rigid templates. 3) Released component maps and specialist reports. 4) Signed structural and fabrication drawings. This example render and all photographs sit outside the dimensional evidence chain.

14. Release checklist

The work is final only when every applicable line has evidence, initials and date.

Gate	Pass requirement	Evidence / sign-off
Parts	R0 pressure/flow; C0 specialist test; G0 condition; FS/FL identity, electrical, direction, bearing and airflow decisions complete	Reports / initials: _____
Actual measurements	D00, A0-D-L/R, B0, S0, CL0, G0-H, R0-H, L0/MR, R0/C0/FS/FL and VEH sheets complete with raw readings and templates	Measurer / owner / date: _____
Drawings	X0/X1/A0/A1 structural release plus G1/G2/R0-E/R3/C1/F1/F2/E1 drawings issued at named revisions	Drawing register: _____
Fabrication	Material certificates/identification, cuts, bends, holes, welds, fasteners, torque provisions, sleeves, drainage and finish access inspected	Fabricator / inspector: _____
Dry fit	One FS + one FL; all direct/local centre checks pass; independent supports/removal; all vehicle, engine, pipe, hose, wire and tool envelopes clear	CL0 and fit sheet: _____
Proof	Released total 2 x MR / 10 minute test passes; critical readings repeated; R3-U remains neutral and both saddles loaded	Structural witness: _____
Finish / assembly	Coating, drainage, hardware, rubbers, hoses, A/C seals/drier and both protected fan branches pass	Finish/torque/electrical sheets: _____
Performance	Cooling pressure/bleed/heat-cycle, A/C evacuation/charge/hot-idle, road/load and first-journey reinspection pass	Commissioning record: _____

Controlled final status

FINAL FIT + FABRICATION + PERFORMANCE RELEASE HOLD until the actual measurement fields, specialist tests, drawings, proof and commissioning records above are complete.

Controlled references

- Technical guide: docs/J40-naturally-aspirated-cooling-pack-restoration-guide-rev-p-20260816.md.
- Fabrication handoff: data/manual/fabrication/na_cooling_connector_arms_rev_p/README.md, measurement_basis.csv, fabricator_cut_list.csv and inspection_checklist.csv.
- Actual image evidence: the controlled front-cooling-stack work-document assets plus the dated project photographs identified in this PDF.
- Superseded and excluded: Rev A 410 mm / 4 mm tall uprights; Rev N X2 adapters; turbo/intercooler/K0 and second-front-fan layouts; every Rev Q provisional envelope estimate.

15. Cost estimate - PKR planning allowance

Budget range at 16 Aug 2026. This is a planning estimate, not a supplier quotation; actual measurements and specialist condition reports control scope.

Baseline scope	Assumption	Low PKR	High PKR
Measurement, rigid templates, drawings and structural review	D00/component capture, LH/RH templates, issued fabrication drawings	15,000	35,000
R0 strip, repair, recore and tests	Reuse sound tanks/headers/necks/rails; new copper/brass core	40,000	80,000
C0 clean/test plus new drier and HNBR seals	Assumes retained condenser passes specialist test	15,000	35,000
G0 repair, mesh/frame finish and edge protection	Retain/re-mesh by actual template	8,000	20,000
FS and FL test/service	Assumes both retained motors/frames remain usable	10,000	25,000
Fabricate all released mounts and carriers	X0/X1/A0/A1/G1/G2/R0-E/R3/C1/F1/F2/E1	45,000	95,000
Blast/prep, 2K primer and satin/semi-gloss finish	Steel parts only; professional 2K controls	15,000	35,000
Cooling service parts	Hoses, clamps, cap, thermostat/seal and coolant	15,000	35,000
Fan electrical parts	Two separate relay/fuse/wire/connector/earth branches	12,000	30,000
A/C evacuation, charge and final specialist work	Excludes major compressor/evaporator repair	10,000	25,000
Baseline subtotal	Before contingency	185,000	415,000
Contingency	15% for hidden damage, rework and price movement	28,000	62,000
PLANNING TOTAL	Hybrid refurbishment, retained major parts passing tests	213,000	477,000

Optional failure-case replacements

Optional addition	Low PKR	High PKR
New matched C0 condenser if retained C0 fails	25,000	50,000
Quality new FS pusher if candidate fails	15,000	45,000
Quality new FL/shroud if retained unit fails	20,000	55,000
Complete bespoke copper/brass R0 instead of recore - incremental	40,000	90,000
Maximum optional increment	100,000	240,000
Worst-case planning envelope	313,000	717,000

Budget assumptions

Ranges include local parts, specialist work and fabrication labour allowances but exclude imported freight/duties, engine-bay repairs outside the cooling pack, compressor/evaporator replacement and sales tax treatment not included by a quoted shop. Obtain three itemised quotations after the actual measurement and retain/replace sheets are signed.

Retail anchors checked 16 Aug 2026: generic 12 V fan PKR 1,999; imported quality 14-inch fan PKR 40,231 before shipping/customs; 2K clear PKR 2,850; 2K primer PKR 12,000; 4 L coolant listings roughly PKR 2,095-4,900. These anchors explain the wide ranges; they are not approved parts.

Recommended procurement route

Recore R0; test and conditionally retain C0, G0, FS and FL; fabricate the mounts from actual templates; buy all service consumables, receiver-drier, seals, isolation, fasteners and wiring new. A complete new pack is sensible only as a bespoke sample-built commission, not an off-the-shelf listing.